

TrES-3b

R-2703

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160616 · created 2026-09-01T00:30:18+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457555.8739 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.017
Transit depth (Rp/Rs)^2	2.21 +/- 0.52 [%]
Orbital Inclination (inc)	84.29 +/- 1.1
Airmass coefficient 1 (a1)	1.176 +/- 0.013
Airmass coefficient 2 (a2)	-0.137 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	None
Optimal Aperture	3.73
Optimal Annulus	12.03
Transit Duration (day)	0.0698 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.149 $\pm$ 0.017 vs 0.16309 (deviation 0.83 σ)
Duration fit vs published	0.0698 d vs 0.0567 d (deviation 2.23 σ)
Mid-time O-C	2.4 min
Depth SNR	8.8
Residual scatter	1.09 %

Light curve

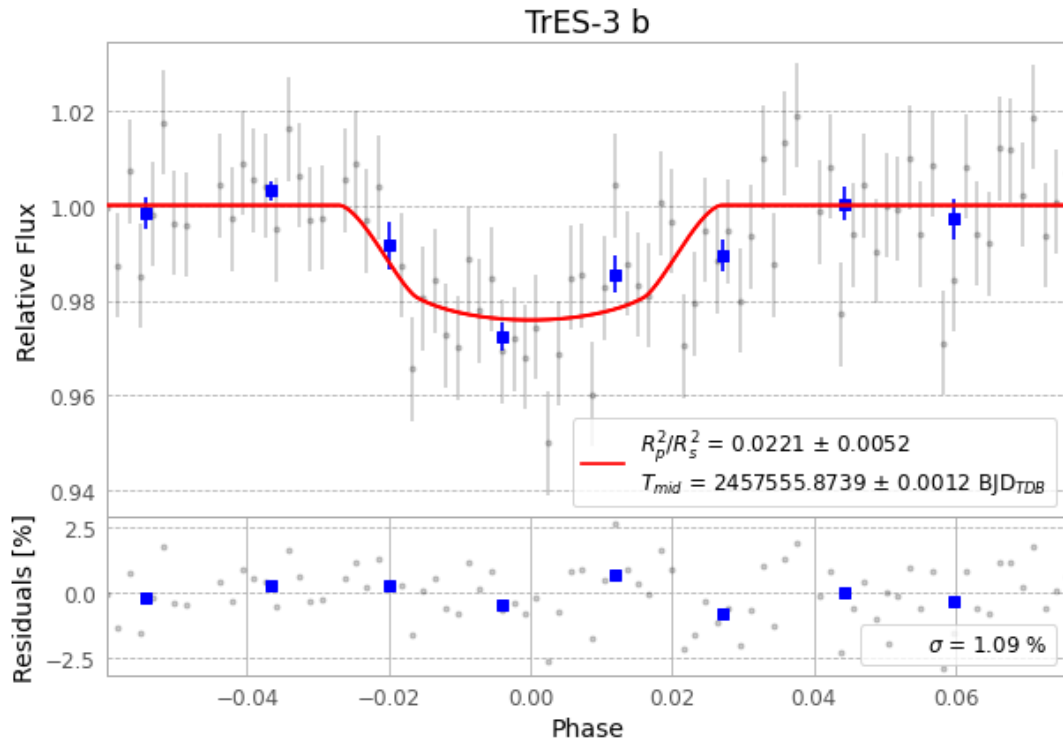


Figure 1 — FinalLightCurve_TrES-3 b_2016-06-16.png (SHA-256 24599ded42e2727e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [136, 240], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3f4e9e83360bcad3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

84 FITS frames (55 MB), TRES-3160616070012.FITS → TRES-3160616111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160616.log (f5e919f07c39...), FinalLightCurve_TrES-3 b_2016-06-16.png (24599ded42e2...), FinalLightCurve_TrES-3 b_2016-06-16.csv (1c5801f6328e...)

Compute resources

Wall time 123.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 00:30Z.